

solve a given problem, while ensuring that no other can crack that code. 'bug-;' is a test of knowledge of compilers. The dare would be to explain a particular error message generated by the compiler compiling bad code. The final hurdle, 'Codessence,' will be a test in reverse-engineering skills, where a given executable has to be reverse-engineered, based on a given set of inputs and outputs. The coder who achieves it by running the executable the least number of times wins. The last date for registration is October 15. You can register at codeprix2005@hotmail.com.

N-GAGE INDIA

Nokia eN-Gages India

On September 14, Nokia launched the N-Gage QD Mobile Gaming Championship, which gives gamers from all over the country to take home upto Rs 10 lakh in cash prizes. The game, *Hero Commando*, specially developed for this championship, will have to be played in bronze and silver rounds, eventually concluding in a gold round on October 31.

Specially designed N-Gage Challenge Buses, each with four N-Gage consoles, will parade through 47 cities inviting players to gun it out at each other and prove their worth.

According to *The Register*, though Nokia has shipped out a million N-Gage QDs to date, about 600,000 units were announced shipped in February of 2004 since October of 2003—the first six months of the QD release—leaving only 400,000 units to be sent out in the next six. This 33 per cent decline in shipments may indicate a waning market for the QD. Even after

reasonably widespread availability of the N-Gage QD in India, since its October 2003 launch, and also its comparatively low price, sales figures for Nokia's



mobile gaming contender are alarmingly low. The championship is perhaps an effort to increase the sales by luring potential mobile gamers to switch over to the QD, and with a Rs 10 lakh bumper prize, it might just do the trick!

For those who want to practice the game before the challenge, download Commando Bronze by sending 260668 as an SMS to 5555, although this requires GPRS connectivity.

HOT NEW SMARTPHONE

Say Hello to Blackberry

Thanks to telecom major Bharti, you will soon be able to drool over the Blackberry—one of the hottest smartphones around,



according to pink newspaper *Business Standard*. The tie-up with Bharti will provide Blackberry customers an Airtel connection and other specialised services. With a price range of Rs 20,000 to Rs

30,000, the target market is likely to be corporate clients in the five big metros. Cheaper models may be introduced soon.

Blackberry smartphones offer several value-added services, including the 'push' technology by which office e-mails can be transferred automatically to the phone. Blackberry's 'SureType' technology allows users to dial or type either by single-handed operation or two-handed thumb typing, without the limitations of traditional phone keypads.

PHONEY MP3 PLAYERS

It Had to Happen

In the August issue of *Digit* we said that by end-2004, phones would double up as MP3 players. We were right! On September 6, Samsung unveiled the clamshell SPH-V5400, touted as the world's first phone with a hard drive. The hard drive stores 1.5 GB, and the phone will retail for \$800 (Rs 37,000) in South Korea—Samsung has not disclosed any plans to sell it outside of that country yet.

The phone can double up as a full-fledged MP3 player, although there are no details on the interface yet.

The idea of phones with hard drives has been with us a long time, so why haven't manufacturers come out with them before? Answer: disk sizes and reliability. Even the arrival of 1-inch drives—which make arming phones with hard disks a possibility—hasn't made production easy or cheap.

Another interesting feature is that the phone can broadcast audio to FM receivers in its vicinity.

What will spur the demand for such handsets is the availability of cheap, small form-factor hard disks. Seagate has developed a 1-inch, 5 GB drive, while Toshiba's drive is even smaller—0.85 in in diameter, and stores 2 to 3 GB. ■



What is a DualDisc?

A DualDisc is a new optical disc format announced by a group of record labels. It is designed to make music CDs more attractive to consumers by using additional content.

How does a DualDisc work?

A DualDisc will have a full CD album on one side and its enhanced version on the other. It will be compatible with any device that plays either CDs or DVDs.

Who is responsible for the DualDisc format?

It is the work of a consortium that includes major labels, such as EMI Music, Sony BMG Music Entertainment, Warner Music Group, and Universal Music Group.

Why do we need DualDiscs?

With DualDisc, musicians will have more than one medium to reach their audience. The DVD side is meant to carry video clips, interviews, concert shows and links to Web sites, in addition to music in the much richer DVD-audio format. It is hoped that this will make the product experience much more immersive and interactive.

When was the format revealed?

The DVD Forum approved the format this June. However, a German company, DVD Plus, claimed a pending patent, which describes a similar product. The patent was filed by DVD Plus chief Dieter Dierks in March 1999.

Where can I find DualDiscs?

DualDiscs have just been announced and are expected to make it to the market by the end of this year.